

Remark 0.1 (The holographic-datum heptagon). The seven presentations of the $\text{SC}^{\text{ch},\text{top}}$ datum are the operadic, Koszul, factorization, BV–BRST, convolution, Drinfeld-centre, and derived-geometric presentations. The Drinfeld-centre face identifies the factorizable representation centre with the representation theory of the chiral derived centre through categorified bar-cobar and half-braiding. The derived-geometric face carries the PTVV 1-shifted symplectic form on $\text{Map}(X \times \mathbb{R}_{\geq 0}, B \text{ SC-Alg})$. All seven presentations factor through the chiral Hochschild cochain complex. The topologisation from $\text{SC}^{\text{ch},\text{top}}$ to E_3^{top} is proved cohomologically for affine Kac–Moody algebras at non-critical level, where the Sugawara conformal vector is explicit. The general chain-level lift for conformal-vector chiral algebras is conjectural; the affine critical level is excluded because Sugawara topologisation is singular there.

THE HOLOGRAPHIC MODULAR KOSZUL DATUM

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ABSTRACT. For the modular-Koszul boundary sector of a 3d holomorphic-topological gauge theory T on $\mathbb{R}_+ \times X$, the comparison data form a seven-component package

$$\mathcal{H}(T) = (\mathcal{A}, \mathcal{A}^i, \mathcal{A}^l, \mathcal{Z}_{\text{ch}}^{\text{der}}(\mathcal{A}), r(z), \Theta_{\mathcal{A}}, \nabla_{\bullet, \bullet}^{\text{hol}}).$$

The entries are the boundary chiral algebra, bar cohomology coalgebra, Verdier-dual Koszul algebra, closed-sector derived centre, spectral r -matrix, universal Maurer–Cartan element, and modular shadow connection. We define each component from the bar complex $\bar{B}(\mathcal{A}) = T^c(s^{-1}\mathcal{A})$ and its single structural identity $D_{\mathcal{A}}^2 = 0$, and compute the datum in three model families: Heisenberg $\mathcal{H}_k$ (abelian Chern–Simons, class **G**), affine $\widehat{\mathfrak{sl}}_{2,k}$ (non-abelian Chern–Simons, class **L**), and Virasoro Vir_c (3d gravity, class **M**). The algebraic closed-sector–boundary–line triangle is recovered on the compact-generator comparison surface; its physical BBL reading requires the open–closed comparison datum. The closed-sector vertex is the chiral derived centre $\mathcal{Z}_{\text{ch}}^{\text{der}}(\mathcal{A})$, the boundary is $\mathcal{A}$ itself, and the line category is modelled by $\mathcal{A}^l\text{-mod}$. The $\text{SC}^{\text{ch},\text{top}}$ structure emerges on the pair $(\mathcal{Z}_{\text{ch}}^{\text{der}}(\mathcal{A}), \mathcal{A})$ of derived centre and boundary algebra. The central organising principle is the $\text{SC}^{\text{ch},\text{top}}$ *heptagon*: seven edges (operadic / Koszul / factorization / BV–BRST / convolution / Drinfeld centre / derived-AG) identify the datum across presentations, with six classical faces and a seventh derived-algebraic-geometry face constructed via PTVV on $\text{Map}(X \times \mathbb{R}_{\geq 0}, B \text{ SC-Alg})$. We construct the anomaly completion: the transgression algebra B_{Θ} with secondary anomaly $u = \eta^2 = \kappa(\mathcal{A}^l) \cdot \omega_g$, whose vanishing at the critical central charge is the Clifford/exterior bifurcation governing the genus tower.

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1. INTRODUCTION

The problem is to organise the algebraic data visible from the modular-Koszul boundary sector of a 3d holomorphic-topological (HT) quantum field theory. A 3d HT theory on the half-space $\mathbb{R}_+ \times X$, where X is a smooth algebraic curve, produces three observation algebras: a boundary chiral algebra $\mathcal{A}$ at $\{0\} \times X$, a physical bulk factorisation algebra in the interior, and a category of topological line operators along $\mathbb{R}_+$. The algebraic engine relating these three is Koszul duality in the homotopical modular chiral realm on algebraic curves: the bar construction on $\mathcal{A}$ produces a coalgebra from which the Koszul dual, the r -matrix, and the genus tower are extracted. The governing identity is $D^2_{\mathcal{A}} = 0$; the governing equation is the Maurer–Cartan equation $D\Theta + \frac{1}{2}[\Theta, \Theta] = 0$.

The holographic modular Koszul datum is

$$\mathcal{H}(T) = (\mathcal{A}, \mathcal{A}^i, \mathcal{A}^!, \mathcal{Z}_{\text{ch}}^{\text{der}}(\mathcal{A}), r(z), \Theta_{\mathcal{A}}, \nabla_{\bullet, \bullet}^{\text{hol}}).$$

It is extracted from a single Maurer–Cartan element by a sequence of projections. The components are:

- (i) the boundary chiral algebra $\mathcal{A}$;
- (ii) for a chosen quadratic presentation $\mathcal{A} = T_{\text{ch}}(V)/(R)$, the presentation coalgebra $\mathcal{A}^i = C_X(s^{-1}V, s^{-2}R)$ and the comparison $q_{\mathcal{A}}: \mathcal{A}^i \rightarrow \bar{B}(\mathcal{A})$;
- (iii) the Verdier bar algebra $K_X(\mathcal{A}) = \mathbb{D}_{\text{Ran}}\bar{B}(\mathcal{A})$ and the chosen chiral Koszul partner $\nu_{\mathcal{A}}: K_X(\mathcal{A}) \rightarrow \mathcal{A}^!$;
- (iv) the closed-string algebra $\mathcal{Z}_{\text{ch}}^{\text{der}}(\mathcal{A})$;
- (v) the classical r -matrix $r(z) = \text{Res}_{0,2}^{\text{coll}}(\Theta_{\mathcal{A}})$;
- (vi) the universal Maurer–Cartan element $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$;
- (vii) the modular shadow connection $\nabla_{g,n}^{\text{hol}} = d - \text{Sh}_{g,n}(\Theta_{\mathcal{A}})$.

The bar complex $\bar{B}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ is an E_1 -chiral coassociative coalgebra: it carries a differential (from the chiral product) and a deconcatenation coproduct (from the tensor coalgebra structure). The $\text{SC}^{\text{ch}, \text{top}}$ structure emerges on the chiral derived centre $\mathcal{Z}_{\text{ch}}^{\text{der}}(\mathcal{A}) = C_{\text{ch}}^{\bullet}(\mathcal{A}, \mathcal{A})$, where the pair $(\mathcal{Z}_{\text{ch}}^{\text{der}}(\mathcal{A}), \mathcal{A})$

is the SC datum with the algebraic closed sector acting on the boundary. A physical bulk action requires the open–closed comparison datum.

Three families illustrate the full range of the datum.

Heisenberg $\mathcal{H}_k$. Abelian Chern–Simons at level k . Shadow class $\mathbf{G}$, shadow depth 2. The r -matrix is $r(z) = k/z$ (vanishing at $k = 0$); the line category is the Fock-charge sector of the abelian theory, collapsing to Vect only after imposing the neutral $k = 0$ quotient; the higher-genus shadow connection has scalar curvature $k \omega_g$. This is the scalar base case.

Affine $\widehat{\mathfrak{sl}}_{2k}$. Non-abelian Chern–Simons at level k . Shadow class $\mathbf{L}$, shadow depth 3. The r -matrix is Yang’s $r(z) = k \Omega_{\text{tr}}/z$ (trace-form; vanishing at $k = 0$). The line category is $\text{Rep}(U_q(\mathfrak{sl}_2))$ at $q = e^{i\pi/(k+2)}$, recovered from the KZ monodromy. The shadow connection is the KZ connection. Every component is matrix-valued.

Virasoro Vir_c . 3d gravity. Shadow class $\mathbf{M}$, infinite depth. The r -matrix is $r_c(z) = (c/2)/z^3 + 2T/z$ (cubic and simple poles); the quartic contact invariant $Q^{\text{contact}} = 10/[c(\zeta c + 22)] \neq 0$. The Koszul dual is Vir_{26-c} , self-dual at $c = 13$. The genus tower is infinite; gravity does not truncate.

The closed-sector–boundary–line algebraic triangle is not an ad hoc diagram. The ordered bar coalgebra is the level-2 twisting input. The boundary-to-closed-sector trace passes through the derived chiral centre, the boundary-to-lines passage uses Koszul duality, and the closed-sector-to-lines edge is the complementarity cofibre. Physical BBL language requires the open–closed comparison datum. Commutativity of the algebraic triangle is Theorem A on the boundary-linear modular-Koszul locus.

The anomaly completion adjoins to the Koszul dual $\mathcal{A}^!$ a degree-1 generator η with Clifford (not exterior) relation $\eta^2 = \kappa(\mathcal{A}^!) \cdot \omega_g$. The resulting transgression algebra B_Θ has a secondary anomaly $u = \eta^2 = \kappa(\mathcal{A}^!) \cdot \omega_g$ that controls the genus tower: when $u \neq 0$ the Clifford algebra is a matrix algebra and genera are Morita-equivalent; when $u = 0$ the handle operators become exterior generators and the theory sees genuine surface topology.

Conventions. Grading is cohomological ($|d| = +1$). Desuspension lowers degree: $|s^{-1}v| = |v| - 1$. The bar propagator is $d \log E(z, w)$, weight 1 in both variables regardless of field weight. OPE-mode notation: $a_{(n)}b$. The augmentation ideal is $\bar{\mathcal{A}} = \ker(\varepsilon)$; the bar complex uses $\bar{\mathcal{A}}$, not $\mathcal{A}$.

2. THE DATUM

Let $\mathcal{A}$ be a chirally Koszul vertex algebra on a smooth projective curve X .

Definition 2.1 (The holographic modular Koszul datum). The *holographic modular Koszul datum* of $\mathcal{A}$ is the seven-component package

$$\mathcal{H}(\mathcal{A}) = (\mathcal{A}, \mathcal{A}^i, \mathcal{A}^!, \mathcal{Z}_{\text{ch}}^{\text{der}}(\mathcal{A}), r(z), \Theta_{\mathcal{A}}, \nabla_{\bullet, \bullet}^{\text{hol}}),$$

with components defined in Definitions 2.2–2.8 below.

2.1. Boundary algebra.

Definition 2.2 (Boundary algebra). The *boundary algebra* is the chiral algebra $\mathcal{A}$ on X , restricted from the holomorphic boundary $\{0\} \times X \subset \mathbb{R}_+ \times X$ of the 3d HT system.

The boundary carries the full OPE data and is the input to the bar construction. In the 3d HT picture, $\mathcal{A}$ is the boundary condition at the $t = 0$ wall: the holomorphic direction produces the OPE, and the topological direction produces the ordering.

2.2. Bar complex. The ordered bar complex is

$$\bar{B}^{\text{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}}),$$

where $\bar{\mathcal{A}} = \ker(\varepsilon)$ is the augmentation ideal and s^{-1} denotes desuspension ($|s^{-1}v| = |v| - 1$). This is an E_1 -chiral coassociative coalgebra: the differential $d_{\bar{B}}$ extracts OPE residues at collision divisors of the Fulton–MacPherson space $\overline{C}_n(\mathbb{C})$, and the deconcatenation coproduct

$$\Delta(a_1 \otimes \cdots \otimes a_n) = \sum_{p=0}^n (a_1 \otimes \cdots \otimes a_p) \otimes (a_{p+1} \otimes \cdots \otimes a_n)$$

splits an ordered tensor sequence at a cut point.

The symmetric bar $\bar{B}^{\Sigma}(\mathcal{A})$ is the Σ_n -coinvariant image under the averaging map $\text{av}: \bar{B}^{\text{ord}} \rightarrow \bar{B}^{\Sigma}$. The descent from ordered to symmetric is R -matrix-twisted: $\bar{B}_n^{\Sigma} = (\bar{B}_n^{\text{ord}})^{R \cdot \Sigma_n}$. The ordered bar is the level-2 chain object before averaging; the symmetric bar is its coinvariant shadow.

Remark 2.3 (SC structure lives on the derived centre). The bar complex $\bar{B}(\mathcal{A})$ is an E_1 -chiral coassociative coalgebra. The $\text{SC}^{\text{ch}, \text{top}}$ structure emerges on the chiral derived centre $\mathcal{Z}_{\text{ch}}^{\text{der}}(\mathcal{A}) = C_{\text{ch}}^{\bullet}(\mathcal{A}, \mathcal{A})$, computed using the bar complex as a resolution. The pair $(\mathcal{Z}_{\text{ch}}^{\text{der}}(\mathcal{A}), \mathcal{A})$ is the SC datum.

2.3. Koszul dual.

Definition 2.4 (Chiral Koszul dual). Fix a quadratic presentation $\mathcal{A} = T_{\text{ch}}(V)/(R)$. Its presentation coalgebra and full-bar comparison are

$$\mathcal{A}^! = C_X(s^{-1}V, s^{-2}R), \quad q_{\mathcal{A}}: \mathcal{A}^! \longrightarrow \bar{B}(\mathcal{A}).$$

Verdier duality forms $K_X(\mathcal{A}) := \mathbb{D}_{\text{Ran}} \bar{B}(\mathcal{A})$ and the dual comparison $\mathbb{D}(q_{\mathcal{A}}): K_X(\mathcal{A}) \rightarrow \mathbb{D}_{\text{Ran}}(\mathcal{A}^!)$. A chiral Koszul-pair datum is a chosen algebra map $\nu_{\mathcal{A}}: K_X(\mathcal{A}) \rightarrow \mathcal{A}^!$, required to be an equivalence on the finite-type Koszul locus.

The passage from $\mathcal{A}$ to $\mathcal{A}^!$ involves a quadratic presentation, the full bar construction, Verdier duality, and a chosen Koszul-pair comparison. On the Koszul locus $q_{\mathcal{A}}$ is a quasi-isomorphism, and $H^{\bullet}(q_{\mathcal{A}})$ identifies the presentation-coalgebra cohomology with bar cohomology. The resulting typed objects and maps are

$$\mathcal{A}^! \xrightarrow{q_{\mathcal{A}}} \bar{B}(\mathcal{A}), \quad K_X(\mathcal{A}) \xrightarrow{\mathbb{D}(q_{\mathcal{A}})} \mathbb{D}_{\text{Ran}}(\mathcal{A}^!), \quad K_X(\mathcal{A}) \xrightarrow{\nu_{\mathcal{A}}} \mathcal{A}^!.$$

The cobar counit $\Omega(\bar{B}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$ records inversion.

2.4. Line-operator category.

Definition 2.5 (Line-operator category). Under the compact-generator open–closed hypotheses, the *line-operator category* is modelled by

$$\mathcal{C} \simeq \mathcal{A}^! \text{-mod}.$$

Objects are topological defect lines along the $\mathbb{R}_+$ -direction; morphisms are junction operators. The braiding on $\mathcal{C}$ comes from crossing in the holomorphic plane: two line operators separated in $\mathbb{C}_z$ can be exchanged by moving one around the other. On the Koszul locus, the r -matrix $r(z)$ determines the braiding explicitly.

2.5. Spectral r -matrix.

Definition 2.6 (Classical r -matrix). The *spectral r -matrix* is the binary genus-0 collision residue

$$r(z) = \text{Res}_{0,2}^{\text{coll}}(\Theta_{\mathcal{A}}) \in \mathcal{A}^! \hat{\otimes} \mathcal{A}^![[z^{-1}, z]].$$

Under $H_{\text{CYBE}}^{\text{rep}}(\mathcal{A}; W)$, the represented kernel $r_{\mathcal{A}} = \rho_{\mathcal{A}}(K_{\mathcal{A}}^{\text{coll}})$ satisfies the classical Yang–Baxter equation. The representation carries the Arnold relation on $\overline{M}_{0,4}$ to CYBE: the three boundary divisors of the genus-0 4-pointed moduli space correspond to channels (12|34), (13|24), (14|23), and evaluating the Arnold relation on $\Omega \otimes \Omega \in (\mathcal{A}^!)^{\otimes 3}$ produces the three terms of $\Sigma[r_{ij}, r_{ik}] = 0$.

The ordered kernel $K_{\mathcal{A}}^{\text{coll}}$ lives in the E_1 convolution algebra $\mathfrak{g}_{\mathcal{A}}^{E_1}$. Its homotopy coinvariant $q_2(K_{\mathcal{A}}^{\text{coll}})$ remains operator-valued. The normalized functional $\tau_{\mathcal{A},2}^{\text{res}}$ gives κ for abelian and scalar families. For non-abelian affine Kac–Moody it gives κ_{dp} , and the Sugawara contribution completes the modular characteristic:

$$\kappa(V_k(\mathfrak{g})) = \tau_{\mathcal{A},2}^{\text{res}} q_2(K_{\mathcal{A}}^{\text{coll}}) + \frac{\dim(\mathfrak{g})}{2} = \frac{\dim(\mathfrak{g})(k + h^\vee)}{2h^\vee},$$

where the $\dim(\mathfrak{g})/2$ term is the simple-pole self-contraction through the adjoint Casimir eigenvalue $2h^\vee$.

2.6. Universal Maurer–Cartan element.

Definition 2.7 (Universal MC element). The *universal Maurer–Cartan element* is

$$\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_o \in \text{MC}(\mathfrak{g}_{\mathcal{A}}^{\text{mod}}),$$

where $D_{\mathcal{A}}$ is the full bar differential and d_o is its linear (genus-0, degree-1) part.

The element $\Theta_{\mathcal{A}}$ is always MC because $D_{\mathcal{A}}^2 = 0$:

$$D\Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0.$$

This is the bar-intrinsic construction: $\Theta_{\mathcal{A}}$ exists for every chirally Koszul algebra, at all degrees, as an inverse-limit $\Theta_{\mathcal{A}} = \varprojlim \Theta_{\mathcal{A}}^{\leq r}$. The modular characteristic, the r -matrix, the shadow connection, and the full genus tower are projections of $\Theta_{\mathcal{A}}$.

2.7. Shadow connection.

Definition 2.8 (Shadow connection). The *modular shadow connection* on $\overline{M}_{g,n}$ is

$$\nabla_{g,n}^{\text{hol}} = d - \text{Sh}_{g,n}(\Theta_{\mathcal{A}}),$$

where $\text{Sh}_{g,n}$ is the shadow projection to the (g, n) -component of the modular convolution algebra.

Flatness of ∇^{hol} is the MC equation projected to (g, n) . At genus 0, $\nabla_{0,n}^{\text{hol}}$ is unconditionally flat; at genus $g \geq 1$, the connection acquires curvature $\kappa(\mathcal{A}) \cdot \omega_g$ (the genus- g anomaly).

The shadow connection encodes the full genus tower. Its monodromy representations are the physical amplitudes: at genus 0, the KZ monodromy for affine algebras; the BPZ monodromy for Virasoro.

2.8. Collision filtration and shadow depth. The modular convolution algebra $\mathfrak{g}_{\mathcal{A}}^{\text{mod}}$ carries a collision filtration $F_{\text{coll}}^0 \supset F_{\text{coll}}^1 \supset \dots$ indexed by minimum pairwise distance on $\overline{C}_n(\mathbb{C})$. Depth 0 is free propagation; depth k is k -fold nested collision. The spectral sequence $E_i^{p,*} = H^*(\text{gr}_{\text{coll}}^p \mathfrak{g}_{\mathcal{A}}^{\text{mod}}, d_o)$ converges to $H^*(\mathfrak{g}_{\mathcal{A}}^{\text{mod}})$. Collapse is governed by shadow depth:

Class	Shadow depth	OPE pole order	Representative
G	2	2	Heisenberg $\mathcal{H}_k$
L	3	3	Affine $\widehat{\mathfrak{g}}_k$
C	4	4	$\beta\gamma$ system
M	∞	≥ 4	Virasoro, $\mathcal{W}_N$

The discriminant $\Delta = 8\kappa S_4$ controls truncation: $\Delta = 0$ (classes **G**, **L**, **C**) gives a finite shadow tower; $\Delta \neq 0$ (class **M**) gives an infinite tower. The discriminant is linear in κ , not quadratic.

3. THE HEISENBERG DATUM

The Heisenberg algebra $\mathcal{H}_k$ at level k is the abelian current algebra generated by a single field J with OPE

$$J(z)J(w) \sim \frac{k}{(z-w)^2}.$$

This is abelian Chern–Simons for $G = U(1)$ at level k .

Proposition 3.1 (Heisenberg datum). *The holographic datum of $\mathcal{H}_k$ is*

$$\mathcal{H}(\mathcal{H}_k) = (\mathcal{H}_k, H^*(\bar{B}(\mathcal{H}_k)), (\text{Sym}^{\text{ch}}(V^*[1]), m_o = -k\omega), \mathcal{Z}_{\text{ch}}^{\text{der}}(\mathcal{H}_k), k/z, k \cdot \eta \otimes \Lambda, d - k\omega_g).$$

Proof. The entries are computed directly from the OPE, the bar differential, and the shadow projection.

(i) *Boundary.* $\mathcal{A} = \mathcal{H}_k$, the free boson at level k .

(ii) *Koszul dual.* $\mathcal{A}^! = \text{Sym}^{\text{ch}}(V^*)$. This is the curved symmetric chiral algebra with $m_o = -k\omega$; it is *not* $\mathcal{H}_{-k}$ (the same algebra has the same κ on both sides, but the duals differ as curved objects).

(iii) *Lines.* $\mathcal{C}$ is the Fock-charge line category of the abelian theory, equivalently the rank-one abelian Yangian/Fock module sector. It collapses to Vect only at $k = 0$ after imposing the neutral quotient.

(iv) *r-matrix.* $r(z) = k/z$. The OPE has a double pole; $d \log$ absorbs one, leaving a simple pole. At $k = 0$: $r = 0$ (abelian limit).

(v) *MC element.* $\Theta_{\mathcal{H}_k} = k \cdot \eta \otimes \Lambda$, where $\eta = d \log(z_1 - z_2)$ is the Arnold form and Λ is the unique (up to scale) bilinear on V . The MC equation is satisfied because $\mathcal{H}_k$ is class **G**: all operations beyond degree 2 vanish.

(vi) *Shadow connection.* $\nabla_{o,n}^{\text{hol}} = d$ at genus 0. At positive genus the shadow connection has scalar curvature $k\omega_g$; it becomes trivial only at $k = 0$ or after cancellation by the dual curvature. $\square$

Remark 3.2 (Modular characteristic). The modular characteristic is $\kappa(\mathcal{H}_k) = k$. This is the abelian prototype: $\tau_{\mathcal{H}_{k,2}}^{\text{res}} q_2(K_{\mathcal{H}_k}^{\text{coll}}) = k = \kappa$. The Koszul complementarity is $\kappa(\mathcal{H}_k) + \kappa(\mathcal{H}_k^!) = 0$ (class **G/L**/free-field complementarity).

Remark 3.3 (Quantised R -matrix). The quantised R -matrix is $R(z) = \exp(k/z)$, scalar and automatically satisfying the Yang–Baxter equation. The monodromy $\exp(-2\pi i k)$ is the Aharonov–Bohm phase.

4. THE KAC–MOODY DATUM

The affine Kac–Moody algebra $\widehat{\mathfrak{g}}_k$ at level k is generated by currents $J^a(z)$ ($a = 1, \dots, \dim(\mathfrak{g})$) with OPE

$$J^a(z)J^b(w) \sim \frac{k\kappa^{ab}}{(z-w)^2} + \frac{f^{ab}_c J^c(w)}{z-w},$$

where κ^{ab} is the Killing form and f^{ab}_c are structure constants. This is 3d Chern–Simons for G at level k . We take $\mathfrak{g} = \mathfrak{sl}_2$ for concreteness; the formulae generalise to arbitrary simple $\mathfrak{g}$.

Proposition 4.1 (Affine $\widehat{\mathfrak{sl}}_2$ datum). *The holographic datum of $\widehat{\mathfrak{sl}}_{2,k}$ is*

$$\mathcal{H}(\widehat{\mathfrak{sl}}_{2,k}) = (\widehat{\mathfrak{sl}}_{2,k}, H^*(\bar{B}(\widehat{\mathfrak{sl}}_{2,k})), \text{CE}^{\text{ch}}(\widehat{\mathfrak{sl}}_{2-k-4}), \mathcal{Z}_{\text{ch}}^{\text{der}}(\widehat{\mathfrak{sl}}_{2,k}), \frac{k\Omega}{z}, \Theta_k, d - \kappa_k \omega_g - \dots),$$

with $q = e^{i\pi/(k+2)}$ and $\kappa_k = \dim(\mathfrak{sl}_2)(k + h^\vee)/(2h^\vee) = 3(k+2)/4$.

Proof. The entries are computed directly from the OPE, the bar differential, and the shadow projection.

(i) *Boundary.* $\mathcal{A} = \widehat{\mathfrak{sl}}_{2k}$, the affine Kac–Moody vertex algebra at level k . The three generators J^a ($a \in \{e, f, h\}$) produce a triple-pole OPE (double pole from the Killing form, simple pole from the Lie bracket).

(ii) *Koszul dual.* $\mathcal{A}^\dagger = \text{CE}^{\text{ch}}(\widehat{\mathfrak{sl}}_{2-k-4})$ after the PBW rectification of the affine bar coalgebra. The level parameter is reflected by $k \mapsto -k - 2h^\vee$, with $h^\vee(\mathfrak{sl}_2) = 2$. The complementarity of scalar modular characteristics is $\kappa(\widehat{\mathfrak{sl}}_{2k}) + \kappa(\widehat{\mathfrak{sl}}_{2-k-4}) = 0$, verified at $k = 0$: $\kappa_0 = 3/2$ and $\kappa_{-4} = -3/2$.

(iii) *Lines.* $C = \mathcal{O}_q^{\text{sh}} \simeq \text{Rep}(U_q(\mathfrak{sl}_2))$. The braided tensor category of line operators is the representation category of the quantum group at $q = e^{i\pi/(k+2)}$. The identification is the affine Drinfeld–Kohno theorem: the KZ monodromy representation on conformal blocks agrees with the quantum-group R -matrix action on tensor products of evaluation modules.

(iv) *r-matrix.* $r(z) = k \Omega_{\text{tr}}/z$ in trace-form convention, where Ω_{tr} is the trace-form Casimir. At $k = 0$: $r = 0$ (abelian limit). In Kazhdan–Lusztig (KZ) normalisation: $r(z) = \Omega_{\text{KZ}}/((k + h^\vee)z)$ with $\Omega_{\text{KZ}} = 2h^\vee \Omega_{\text{tr}}$ the Killing-normalised Casimir. The two conventions are related by a *level-dependent rescaling of the Casimir*, not by the naive identity $k \Omega_{\text{tr}} = \Omega_{\text{KZ}}/(k + h^\vee)$: the latter fails as a rational-function equation in k (at $k = 0$ the left side vanishes, the right side is finite and non-zero). The correct statement is that r_{tr} and r_{KZ} are two presentations of the same underlying operadic datum in different bases, exchanged by $\Omega \mapsto \Omega$ and $k \mapsto k/((k + h^\vee) \cdot 2h^\vee)$. At $k = -h^\vee = -2$: trace-form finite, KZ diverges (Sugawara singularity at critical level).

(v) *MC element.* $\Theta_k = \Theta_{\widehat{\mathfrak{sl}}_{2k}}$ is determined by $D_{\mathcal{A}}^2 = 0$. The shadow tower terminates at depth 3 (class **L**): the quartic invariant $S_4 = 0$ (the discriminant $\Delta = 8\kappa S_4 = 0$), so the shadow obstruction tower is finite.

(vi) *Shadow connection.* At genus 0, degree n , the shadow connection is the KZ connection:

$$\nabla_{0,n}^{\text{hol}} = d - \sum_{i < j} r_{ij} dz_{ij} = d - \frac{1}{k + h^\vee} \sum_{i < j} \frac{\Omega_{ij} dz_{ij}}{z_{ij}},$$

where $dz_{ij} = d(z_i - z_j)$. Flatness of the KZ connection is the MC equation at genus 0, degree n . At genus $g \geq 1$, the connection acquires curvature $\kappa_k \cdot \omega_g$ with $\kappa_k = 3(k+2)/4$. $\square$

Remark 4.2 (Quantum group line category). The identification $C \simeq \text{Rep}(U_q(\mathfrak{sl}_2))$ is a reconstruction theorem: the monodromy of the genus-0 shadow connection $\nabla_{0,n}^{\text{hol}}$ (the KZ connection) generates a braided tensor category that is equivalent to the representation category of the quantum group. The spectral parameter z in $r^{\text{coll}}(z) = k \Omega_{\text{tr}}/z$ is the trace-form collision parameter. The KZ conformal-block normalization uses $\Omega_{\text{KZ}}/((k + h^\vee)z)$. On evaluation modules, the same spectral variable becomes the additive spectral parameter of the Yangian. The full quantised R -matrix $R(z) = 1 + r(z) + O(\hbar^2)$ satisfies the quantum Yang–Baxter equation.

Remark 4.3 (Critical level). At $k = -h^\vee = -2$, the KZ normalization is singular and $\kappa = 0$ on the affine scalar line. The trace-form current OPE does not vanish. The Feigin–Frenkel centre replaces the non-critical Sugawara construction, and the critical Gaudin algebra is recovered from the centre rather than from a regular value of the KZ connection. The degree-2 ordered chiral homology $H_{\text{ord},2}^1$ doubles from 4 to 8 for $\mathfrak{sl}_2$ (finite computation on $V \otimes V$, $\chi_{\text{ord},2} = -4$ conserved), and Koszulness fails for the full symmetric bar, whose cohomology becomes $\Omega^*(\text{Op}_{\mathfrak{sl}_2}(D))$ (infinite-dimensional; Frenkel–Teleman

2006; distinct object from the degree-2 ordered slice). The Feigin–Frenkel centre is the algebra of $\mathfrak{sl}_2$ -opers on the formal disk: $Z(V_{-2}(\mathfrak{sl}_2)) \simeq \text{Fun}(\text{Op}_{\mathfrak{sl}_2}(D))$. The shadow connection degenerates, and the line category jumps from semisimple to abelian.

5. THE VIRASORO DATUM

The Virasoro algebra Vir_c at central charge c is generated by the stress tensor T with OPE

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}.$$

This is 3d gravity: $SL(2, \mathbb{R}) \times SL(2, \mathbb{R})$ Chern–Simons at levels $k_{\pm} = \ell/(4G)$, with $c = 3\ell/(2G)$ (Brown–Henneaux).

Proposition 5.1 (Virasoro datum). *The holographic datum of Vir_c is*

$$\mathcal{H}(\text{Vir}_c) = (\text{Vir}_c, H^*(\bar{B}(\text{Vir}_c)), \text{Vir}_{26-c}, \mathcal{Z}_{\text{ch}}^{\text{der}}(\text{Vir}_c), r_c(z), \Theta_c, \nabla_c^{\text{hol}}),$$

with

$$\begin{aligned} r_c(z) &= \frac{c/2}{z^3} + \frac{2T}{z}, \\ \kappa_c &= \frac{c}{2}, \\ \nabla_{\text{I}, \text{I}}^{\text{hol}} &= d - \kappa_c \omega_1 - \frac{\text{IO}}{c(5c+22)} \delta_4 - \dots \end{aligned} \quad (5.1)$$

Proof. The entries are computed directly from the OPE, the bar differential, and the shadow projection.

(i) *Boundary.* $\mathcal{A} = \text{Vir}_c$, the Virasoro vertex algebra.

(ii) *Koszul dual.* $\mathcal{A}^! = \text{Vir}_{26-c}$. The Koszul involution is $c \mapsto 26 - c$. The complementarity sum is $\kappa(\text{Vir}_c) + \kappa(\text{Vir}_{26-c}) = c/2 + (26 - c)/2 = 13$. Self-dual at $c = 13$ (not $c = 26$).

(iii) *Lines.* C_c is the line-operator category, determined by the r -matrix monodromy. For the Virasoro algebra, the line category is infinite-depth: the monodromy representation is determined by the full BPZ differential equations, not by a finite-dimensional quantum group.

(iv) *r -matrix.* The r -matrix $r_c(z) = (c/2)/z^3 + 2T/z$ has a cubic pole (gravitational, from the quartic OPE pole after d log-absorption) and a simple pole (stress-tensor exchange). The quartic contact invariant

$$Q^{\text{contact}} = \frac{\text{IO}}{c(5c+22)} \neq 0$$

is nonzero for all $c \neq 0, c \neq -22/5$; this is the obstruction to extending $r(z)$ from degree 2 to degree 4, and its non-vanishing is the signature of class **M**.

(v) *MC element.* $\Theta_c = \Theta_{\text{Vir}_c}$ is the infinite-depth MC element. The shadow tower does not truncate: all shadow invariants S_r ($r \geq 2$) are nonzero. The modular characteristic is $\kappa(\text{Vir}_c) = S_2 = c/2$.

(vi) *Shadow connection.* The shadow connection (5.1) has leading curvature $\kappa_c \omega_g = (c/2) \omega_g$ at genus g , plus an infinite series of corrections from the shadow obstruction tower. $\square$

Remark 5.2 (The gravitational hierarchy). Shadow class **M** means the A_{∞} tower $(m_1, m_2, m_3, \dots)$ is infinite: every higher operation m_k is nonzero. The generating depth is $d_{\text{gen}} = 3$ (the operation m_3 determines all m_k recursively via the MC equation), but the algebraic depth is $d_{\text{alg}} = \infty$ (no operation vanishes). Gravity does not truncate.

Remark 5.3 (Two critical values of c). The two critical central charges play distinct roles:

- (i) $c = 13$: Koszul self-duality ($\text{Vir}_{13}^! = \text{Vir}_{13}$). The complementarity parameter is $\kappa + \kappa' = 13 \neq 0$; self-duality is a symmetry enhancement within the perturbative regime.
- (ii) $c = 26$: anomaly cancellation ($\kappa(\text{Vir}_{26-c}) = (26 - c)/2 = 0$ at $c = 26$). The Clifford/exterior bifurcation (Section 8). This is the bosonic string.

Self-duality and anomaly cancellation are separated by 13 units of central charge.

Remark 5.4 (Primitive coproduct). The transferred coproduct Δ_z^{Vir} is strictly primitive at all degrees: the ghost-number obstruction intrinsic to Drinfeld–Sokolov reduction kills every higher-product-lattice tree correction. All gravitational dynamics resides in the collision residue and the infinite $\mathcal{A}_\infty$ product tower; the coproduct channel is identically zero.

6. COMPARISON TABLE

Component	$\mathcal{H}_k$	$\widehat{\mathfrak{sl}}_{2k}$	Vir_c
$\mathcal{A}$ (boundary)	$\mathcal{H}_k$	$\widehat{\mathfrak{sl}}_{2k}$	Vir_c
$\mathcal{A}^!$ (Koszul dual)	$\text{Sym}^{\text{ch}}(V^*)$	$\text{CE}^{\text{ch}}(\widehat{\mathfrak{sl}}_{2-k-4})$	Vir_{26-c}
C (lines)	Fock charges	$\text{Rep}(U_q(\mathfrak{sl}_2))$	C_c (BPZ)
$r(z)$	k/z	$k \Omega_{\text{tr}}/z$	$(c/2)/z^3 + 2T/z$
κ	k	$3(k+2)/4$	$c/2$
$\kappa + \kappa'$	0	0	13
Class	G	L	M
Depth	2	3	∞
Q^{contact}	0	0	$10/[c(5c+22)]$
∇^{hol}	d (trivial)	KZ	BPZ (infinite)
Physics	abelian CS	non-abelian CS	3d gravity

The three families represent three qualitatively distinct regimes. Class **G** is Gaussian: scalar, trivial lines, the datum collapses to a single number k . Class **L** is Lie-theoretic: matrix-valued, finite quantum-group lines, the datum is determined by $\mathfrak{g}$ and k . Class **M** is gravitational: infinite depth, no finite quantum group, the datum requires the full infinite tower Θ_c .

7. THE CLOSED-SECTOR–BOUNDARY–LINE TRIANGLE

The closed-sector–boundary–line triangle is the algebraic comparison surface underlying the physical BBL diagram of a 3d holomorphic-topological gauge theory. The ordered bar complex is the common source: its trace produces the boundary-to-closed-sector map, its Verdier dual produces the boundary-to-lines map, and its complementarity cofibre produces the closed-sector-to-lines map. A physical bulk vertex is obtained only after the open–closed comparison datum is supplied.

7.1. The three vertices.

Definition 7.1 (Closed-sector–boundary–line triangle). Let $\mathcal{A}$ be a chirally Koszul vertex algebra. The *closed-sector–boundary–line triangle* of $\mathcal{A}$ consists of:

- (i) *Boundary*: the chiral algebra $\mathcal{A}$ itself, the input to the bar construction.

(ii) *Closed sector*: the chiral derived centre

$$\mathcal{A}_{\text{cl}} \simeq \mathcal{Z}_{\text{ch}}^{\text{der}}(\mathcal{A}) = C_{\text{ch}}^{\bullet}(\mathcal{A}, \mathcal{A}),$$

computed by chiral Hochschild cochains (not topological Hochschild; the geometry of the curve determines which chiral/topological/categorical Hochschild theory applies). A physical bulk interpretation requires the open–closed comparison datum.

(iii) *Lines*: under compact-generation and line-comparison hypotheses, the category $\mathcal{C} \simeq \mathcal{A}^1\text{-mod}$ of modules over the chiral Koszul dual.

The $\text{SC}^{\text{ch}, \text{top}}$ structure lives on the *pair* $(\mathcal{Z}_{\text{ch}}^{\text{der}}(\mathcal{A}), \mathcal{A})$: the chiral derived centre carries brace operations and a chiral Gerstenhaber bracket, and the boundary algebra $\mathcal{A}$ is acted upon by the algebraic closed sector via the universal brace. The SC datum is two-coloured with directionality: closed-sector-to-boundary only, no boundary-to-closed-sector map. A physical bulk-to-boundary reading requires the open–closed comparison datum.

7.2. The three edges.

(i) *Boundary* $\rightarrow$ *closed sector* (the bar map). The bar construction maps $\mathcal{A}$ to $\bar{B}(\mathcal{A})$; the trace map $\text{tr}_g : \bar{B}^{(g)}(\mathcal{A}) \rightarrow \mathbf{C}_g(\mathcal{A})$ projects the bar coalgebra into the ambient complex $\mathbf{C}_g(\mathcal{A}) = R\Gamma(\bar{\mathcal{M}}_g, \mathcal{Z}(\mathcal{A}))$. The image $\mathbf{Q}_g(\mathcal{A}) = \text{im}(\text{tr}_g)$ is one half of the complementarity decomposition.

(ii) *Closed sector* $\rightarrow$ *lines* (the complementarity cofibre). The complementarity theorem gives

$$\mathbf{C}_g(\mathcal{A}) \simeq \mathbf{Q}_g(\mathcal{A}) \oplus \mathbf{Q}_g(\mathcal{A}^1).$$

The cofibre $\text{cofib}(\mathbf{Q}_g(\mathcal{A}) \hookrightarrow \mathbf{C}_g(\mathcal{A})) \simeq \mathbf{Q}_g(\mathcal{A}^1)$ is the dual shadow, controlling the line-operator category through the centre–module adjunction.

(iii) *Boundary* $\rightarrow$ *Lines* (Koszul duality). The full bar coalgebra gives $K_X(\mathcal{A}) = \mathbb{D}_{\text{Ran}} \bar{B}(\mathcal{A})$, and the chosen pair map $\nu_{\mathcal{A}} : K_X(\mathcal{A}) \rightarrow \mathcal{A}^1$ gives the line algebra. On a quadratic Koszul chart, $H^{\bullet}(q_{\mathcal{A}})$ and $\mathbb{D}(q_{\mathcal{A}})$ compare this construction with the presentation coalgebra $\mathcal{A}^1$. The module category $\mathcal{A}^1\text{-mod}$ is the categorical output.

Commutativity of the triangle is Theorem A on the boundary-linear modular-Koszul locus.

7.3. Reconstruction from the datum. The datum $\mathcal{H}(\mathcal{A})$ is complete on the boundary-linear modular-Koszul comparison surface.

Proposition 7.2 (Recovery theorems). *The following reconstructions hold:*

- (R1) Collision residue is twisting. $\text{Res}_{0,2}^{\text{coll}}(\Theta_{\mathcal{A}}) = \tau|_{\beta^2(\mathcal{A})}$ (the universal twisting morphism restricted to bar degree 2).
- (R2) CYBE from Arnold. The three boundary divisors of $\bar{\mathcal{M}}_{0,4}$ produce the three terms of $\Sigma[r_{ij}, r_{ik}] = 0$.
- (R3) Affine shadow = KZ. For $\mathcal{A} = \widehat{\mathfrak{g}}_k$: $\text{Sh}_{0,n}(\Theta_{\widehat{\mathfrak{g}}_k}) = (k + h^{\vee})^{-1} \sum_{i < j} \Omega_{ij} dz_{ij} / z_{ij}$.
- (R4) Quartic as L_{∞} obstruction. $[\mathfrak{R}_4^{\text{mod}}(\mathcal{A})] = [o_4(\mathcal{A})] \in H^2(\mathcal{A}_{4,0}^{\text{sh}})$ is the obstruction to extending $r(z)$ from degree 2 to degree 4. Vanishes for $\mathbf{G}, \mathbf{L}, \mathbf{C}$; nonzero for $\mathbf{M}$.

Remark 7.3 (BBL for Heisenberg). Boundary = $\mathcal{H}_k$. Closed sector = $\mathcal{Z}_{\text{ch}}^{\text{der}}(\mathcal{H}_k)$. Bakalov–De Sole–Kac compute the bounded rank-one even superboson dimensions as $(2, 1, 0, \dots)$; a bounded-to-chart quasi-isomorphism transports this vector to $\mathcal{Z}_{\text{ch}}^{\text{der}}(\mathcal{H}_k)$ [2, Theorem 7.4]. Lines are the abelian Fock-charge sector, with a neutral Vect quotient at $k = 0$. The triangle degenerates only after this neutral quotient; for generic k the scalar curvature records the abelian monodromy.

Remark 7.4 (BBL for Kac–Moody). Boundary = $\widehat{\mathfrak{sl}}_{2k}$. Closed sector = $\mathcal{Z}_{\text{ch}}^{\text{der}}(\widehat{\mathfrak{sl}}_{2k})$, whose support is computed by the family datum $H_H(V_k(\mathfrak{sl}_2); S_{V_k(\mathfrak{sl}_2)})$. The adjoint $\mathfrak{sl}_2$ belongs to the zero-mode derivation prequotient. The bounded affine groups satisfy the Bakalov–De Sole–Kac conjectural exterior-power bounds; their passage to the curve chart uses a bounded-to-chart quasi-isomorphism. Lines = $\text{Rep}(U_q(\mathfrak{sl}_2))$. The Drinfeld–Kohno theorem makes the boundary-to-lines edge constructive: the monodromy of the KZ connection generates $\text{Rep}(U_q(\mathfrak{sl}_2))$. The closed-sector-to-lines edge is the complementarity cofibre: $\mathbf{Q}_g(\widehat{\mathfrak{sl}}_{2-k-4})$ controls the line-operator anomaly.

Remark 7.5 (BBL for Virasoro). Boundary = Vir_c . Closed sector = $\mathcal{Z}_{\text{ch}}^{\text{der}}(\text{Vir}_c)$. Its bounded vertex cohomology has one-dimensional groups in degrees $\{0, 2, 3\}$ [2, Theorem 7.2]; a bounded-to-chart quasi-isomorphism transports this support to the chiral curve chart. Lines = C_c : the infinite-depth line category, governed by the BPZ equations. The triangle is fully active at all three algebraic vertices; the closed-sector-to-lines edge is controlled by the Virasoro complementarity $\mathbf{Q}_g(\text{Vir}_c) \oplus \mathbf{Q}_g(\text{Vir}_{26-c})$.

8. THE ANOMALY COMPLETION

8.1. The transgression algebra. Fix a chirally Koszul algebra $\mathcal{A}$ with Koszul dual $B = \mathcal{A}^\dagger$. The curvature of the bar complex on a genus- g curve is $d_{\text{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g$. The Koszul dual carries dual curvature $\lambda = \kappa(B) \cdot \omega_g$.

Definition 8.1 (Transgression algebra). The *transgression algebra* is

$$B_\lambda = B\langle \eta \rangle / (\eta^2 - \lambda), \quad |\eta| = 1, \quad d_{B_\lambda}(\eta) = 0,$$

where η is a degree-1 generator with Clifford (not exterior) relation $\eta^2 = \lambda = \kappa(B) \cdot \omega_g$, and the differential on B is extended by $d_{B_\lambda}(b) = d_B(b) + [\eta, b]$ for $b \in B$.

The relation $\eta^2 = \lambda$ is Clifford when $\lambda \neq 0$ and exterior when $\lambda = 0$. This is the algebraic encoding of the anomaly: $\lambda = 0$ means the theory is anomaly-free, and the algebra structure changes qualitatively.

8.2. The secondary anomaly.

Definition 8.2 (Secondary anomaly). The *secondary anomaly* is

$$u = \eta^2 = \lambda = \kappa(\mathcal{A}^\dagger) \cdot \omega_g.$$

It is closed, central, and acts as the scalar $\kappa(\mathcal{A}^\dagger)$ on standard modules.

The secondary anomaly is the bifurcation parameter for the genus tower. Its behaviour depends on the Koszul complementarity:

Family	$\kappa(\mathcal{A}^\dagger)$	u	Regime
$\mathcal{H}_k$	$-k$	$-k \omega_g$	$u = 0$ at $k = 0$
$\widehat{\mathfrak{sl}}_{2k}$	$3(-k-2)/4$	$3(-k-2)\omega_g/4$	$u = 0$ at $k = -2$ (critical)
Vir_c	$(26-c)/2$	$(26-c)\omega_g/2$	$u = 0$ at $c = 26$ (string)

8.3. Genus- g Clifford completion.

Construction 8.3 (Clifford completion). At genus g , adjoin $2g$ handle operators $\alpha_1, \beta_1, \dots, \alpha_g, \beta_g$ with

$$\alpha_i \beta_j + \beta_j \alpha_i = \delta_{ij} u, \quad \alpha_i \alpha_j + \alpha_j \alpha_i = 0, \quad \beta_i \beta_j + \beta_j \beta_i = 0.$$

The genus- g Clifford completion is the algebra

$$\mathfrak{G}_g(\mathcal{A}) = B_\lambda \otimes \text{Cl}(H_1(\Sigma_g); u).$$

The handle operators record the genus- g topology. Their anticommutator is controlled by the secondary anomaly u , which splits quantum field theory into two qualitative regimes.

8.4. The dichotomy.

Proposition 8.4 (Clifford/exterior dichotomy). *The genus tower of $\mathcal{A}$ bifurcates at $u = 0$:*

- (i) *When $u \neq 0$: the Clifford algebra $\text{Cl}(H_1(\Sigma_g); u)$ is a matrix algebra $M_{2^g}(\mathbb{C})$, of vector-space dimension 2^{2^g} , after inverting u . Genus- g is Morita equivalent to genus-0. All genus corrections are resumable from genus-0 data. This is the perturbative regime.*
- (ii) *When $u = 0$: the Clifford anticommutator vanishes. The handle operators become exterior generators on $H_1(\Sigma_g)$ (dimension 2^{2^g}). The theory sees genuine surface topology. The genus tower collapses ($\kappa_{\text{eff}} = 0$).*

Proof. Standard Clifford algebra theory. When the bilinear form $u \cdot \langle \cdot, \cdot \rangle$ on $H_1(\Sigma_g, \mathbb{Z})$ is nondegenerate (i.e., $u \neq 0$), the Clifford algebra is a matrix algebra $M_{2^g}(\mathbb{C})$ (even-dimensional, nondegenerate symplectic). When $u = 0$, the bilinear form degenerates and the Clifford algebra becomes the exterior algebra $\Lambda^*(H_1(\Sigma_g)^*)$. $\square$

8.5. The three families. *Heisenberg.* $u = -k \omega_g$. At $k = 0$: $u = 0$, exterior regime. At generic k : $u \neq 0$, perturbative. The bifurcation is at the abelian limit.

Kac–Moody. $u = 3(-k - 2)\omega_g/4$. At $k = -2$ (critical level): $u = 0$, the Feigin–Frenkel centre appears, the genus tower collapses. At generic k : $u \neq 0$, perturbative. The critical level is the bifurcation point.

Virasoro. $u = (26 - c)\omega_g/2$. At $c = 26$: $u = 0$, the bosonic string matter contribution cancels the bc -ghost anomaly in the combined matter-ghost system. The pure Virasoro gravity partition function still has its own collision tower. At generic c : $u \neq 0$, perturbative. At $c = 13$: $u = 13\omega_g/2 \neq 0$, so Koszul self-duality lives within the perturbative regime.

The gravitational partition function at genus g is the super-trace of the MC element through the Clifford factor:

$$Z_g = \text{str}_{\mathfrak{g}_g/B_\Theta}(\Theta_c^{(g)}).$$

The complementarity theorem gives a Lagrangian decomposition

$$H^*(\overline{\mathcal{M}}_g, \mathcal{Z}(\text{Vir}_c)) \simeq \mathbf{Q}_g(\text{Vir}_c) \oplus \mathbf{Q}_g(\text{Vir}_{26-c})$$

into boundary (+1-eigenspace of Verdier involution) and bulk (−1-eigenspace) sectors. The shifted symplectic structure makes both Lagrangian.

8.6. The anomaly parameter as holographic observable. The secondary anomaly u is the single scalar that governs the passage from genus 0 to all genera. For each family, u vanishes at a distinguished value of the level/central charge:

Family	$u = 0$ at	Physical meaning	Regime
$\mathcal{H}_k$	$k = 0$	trivial theory	exterior
$\widehat{\mathfrak{g}}_k$	$k = -h^\vee$	critical level, Feigin–Frenkel	exterior
Vir_c	$c = 26$	bosonic string, anomaly cancellation	exterior

The datum $\mathcal{H}(\mathcal{A})$ together with the anomaly completion B_Θ controls the modular-Koszul boundary-linear sector of the 3d HT correspondence: the seven components determine the genus-0 comparison data, and the secondary anomaly u governs the genus tower inside this sector. The entire structure descends from a single identity: $D_{\mathcal{A}}^2 = 0$.

9. CONCLUDING REMARKS

9.1. **The E_n circle.** The holographic datum sits inside the E_n operadic circle:

$$\begin{aligned} E_3^{\text{top}}(\text{physical bulk}) &\rightarrow E_2(\text{boundary chiral}) \rightarrow E_1(\text{bar/QG}) \\ &\rightarrow E_2(\text{Drinfeld centre}) \rightarrow E_3^{\text{top}}(\text{derived centre}). \end{aligned}$$

The arrows are restriction to codimension-2 defect, ordered bar complex, Drinfeld-centre passage to categorical half-braidings, and higher Deligne passage to the derived centre. The Drinfeld centre is the right adjoint to the forgetful functor from braided to monoidal categories; the averaging map is the later lossy $E_2 \rightarrow E_\infty$ shadow. For simple $\mathfrak{g}$, the E_3 -topological deformation space of $\mathcal{Z}_{\text{ch}}^{\text{der}}(V_k(\mathfrak{g}))$ is $H^3(\mathfrak{g})[[\hbar]]$, one-dimensional; the formal disk restriction recovers the Costello–Francis–Gwilliam Chern–Simons E_3 -algebra. The passage from $\text{SC}^{\text{ch},\text{top}}$ to E_3 requires a conformal vector at non-critical level together with a BRST antighost contraction; for affine Kac–Moody this is the Sugawara primitive. This datum makes $\mathbb{C}$ -translations Q -exact and topologises the holomorphic direction. Without such a primitive the structure remains $\text{SC}^{\text{ch},\text{top}}$.

9.2. **Scope.** The topologisation $\text{SC}^{\text{ch},\text{top}} \rightarrow E_3^{\text{top}}$ is proved for affine Kac–Moody algebras $V_k(\mathfrak{g})$ at non-critical level $k \neq -h^\vee$ (where the Sugawara construction is explicit). For general chiral algebras with conformal vector (Virasoro, $\mathcal{W}$ -algebras), the topologisation is cohomological but the chain-level statement is open. The proof is cohomological: for class **M** algebras, where chain-level data is essential, E_3 may exist only on cohomology.

For a finite set $S_{\mathcal{A}}$, Theorem H identifies the curve-chart cohomology with the support model in a family datum $H_H(\mathcal{A}; S_{\mathcal{A}})$. The bounded vertex complex enters through a bounded-to-chart quasi-isomorphism. The Gelfand–Fuchs cohomology of the mode Lie algebra is a separate functor and carries its own support.

9.3. **From the datum to physics.** The derived chiral centre is the universal closed-sector algebra on the boundary-linear comparison surface of a 3d HT gauge theory. Under the BBL identification it is compared with the physical bulk observables: the bar construction on $\mathcal{A}$ produces a coalgebra; the chiral Hochschild cochain complex of $\mathcal{A}$ is computed from the bar resolution; the $\text{SC}^{\text{ch},\text{top}}$ structure on the pair $(\mathcal{Z}_{\text{ch}}^{\text{der}}(\mathcal{A}), \mathcal{A})$ encodes the open/closed interaction. The datum $\mathcal{H}(\mathcal{A})$ is the compact package for this comparison surface.

Three boundaries remain. The global BBL triangle is proved in the boundary-linear direction, while full reconstruction is open. The class **M** topologisation is cohomological until a chain-level model is built. The E_n circle closes only where the Drinfeld-centre identification

$$Z(\mathbf{U}_{\mathcal{A}}) \simeq \mathcal{Z}_{\text{ch}}^{\text{der}}(\mathcal{A})$$

has been proved.

REFERENCES

- [1] A. Beilinson and V. Drinfeld, *Chiral Algebras*, AMS Colloquium Publications **51**, 2004.
- [2] B. Bakalov, A. De Sole, and V. G. Kac, *Computation of cohomology of vertex algebras*, Japan J. Math. **16** (2021), 81–154.
- [3] K. Costello and O. Gwilliam, *Factorization Algebras in Quantum Field Theory*, Vols. 1–2, Cambridge University Press, 2017–2021.
- [4] K. Costello, D. Francis, and D. Gwilliam, *Chern–Simons factorization algebras and knot polynomials*, arXiv:2602.12412, 2026.
- [5] K. Costello and D. Gaiotto, Vertex algebras and 4d $\mathcal{N} = 2$ theories, *JHEP* **2019**(06):018, 2019.
- [6] V. Drinfeld and D. Kazhdan, On the structure of quantum groups, *Funct. Anal. Appl.* **23**(1989), 115–132.

- [7] P. Etingof and A. Varchenko, Geometry and classification of solutions of the classical dynamical Yang–Baxter equation, *Comm. Math. Phys.* **192**(1998), 77–120.
- [8] B. Feigin and E. Frenkel, Affine Kac–Moody algebras at the critical level and Gelfand–Dikii algebras, *Int. J. Mod. Phys. A* **7**(Suppl. 1A)(1992), 197–215.
- [9] D. Gaiotto and K. Zeng, Interface minimal model holography and topological string theory, arXiv:2603.08783, 2026.
- [10] V. Ginzburg and M. Kapranov, Koszul duality for operads, *Duke Math. J.* **76**(1994), 203–272.
- [11] A. Z. Khan and K. Zeng, Poisson vertex algebras and three-dimensional gauge theory, arXiv:2502.13227, 2025.
- [12] V. Knizhnik and A. Zamolodchikov, Current algebra and Wess–Zumino model in two dimensions, *Nuclear Phys. B* **247**(1984), 83–103.
- [13] M. Livernet, Koszulity of the category of Swiss-cheese operads, *J. Pure Appl. Algebra* **204**(2006), 475–495.
- [14] J.-L. Loday and B. Vallette, *Algebraic Operads*, Grundlehren **346**, Springer, 2012.
- [15] R. Lorgat, *Modular Koszul Duality*, Vols. I–III, 2026.